

45 Days Summer Training Program

Assignment: 2

1. Create a tuple: my_tuple = (10, 20, 30, 40, 50).

Print:

- First element
- Last element
- Length of the tuple
- Elements from index 1 to 3

➤ my_tuple = (10, 20, 30, 40, 50)

```
print("First element:", my_tuple[0])
print("Last element:", my_tuple[-1])
print("Length of the tuple:", len(my_tuple))
print("Elements from index 1 to 3:", my_tuple[1:4])
```

- output:-

First element: 10

Last element: 50

Length of the tuple: 5

Elements from index 1 to 3: (20, 30, 40)

2. Create a tuple:

```
fruits = ("apple", "banana", "mango", "orange")
```

Print:

- Second fruit
- Last two fruits
- Total number of fruits

➤ fruits = ("apple", "banana", "mango", "orange")

```
print("Second fruit:", fruits[1])
print("Last two fruits:", fruits[-2:])
print("Total number of fruits:", len(fruits))
```

- output:-

Second fruit: banana

Last two fruits: ('mango', 'orange')

Total number of fruits: 4

3. Create a set: numbers = {10, 20, 30, 40, 50}

Print:

- Complete set
- Length of the set
- Check whether 30 is present in the set

```
➤ numbers = {10, 20, 30, 40, 50}
print("Complete set:", numbers)

print("Length of the set:", len(numbers))

print("30 is present in the set:", 30 in numbers)
```

- output :-

Complete set: {40, 10, 50, 20, 30}

Length of the set: 5

30 is present in the set: True

4. Create two sets:

set1 = {1, 2, 3, 4}

set2 = {3, 4, 5, 6}

Find:

- Union
- Intersection

```
➤ set1 = {1, 2, 3, 4}
set2 = {3, 4, 5, 6}

print("Union:", set1.union(set2))
print("Intersection:", set1.intersection(set2))
```

- output :-

Union: {1, 2, 3, 4, 5, 6}

Intersection: {3, 4}

5. Create a dictionary:

```
student = {"name": "Kriti", "age": 20, "course": "Python"}
```

Print:

- Complete dictionary
- Student name
- Student age

```
➤ student = {"name": "Kriti", "age": 20, "course": "Python"}
```

```
print("Complete dictionary:", student)
print("Student name:", student["name"])
print("Student age:", student["age"])
print("Student course:", student["course"])
```

- output :-

```
Complete dictionary: {'name': 'Kriti', 'age': 20, 'course': 'Python'}
Student name: Kriti
Student age: 20
Student course: Python
```

❖ Student course Create a list:

```
numbers = [12, 45, 7, 23, 56, 89, 34]
```

Write a program to find:

- o Largest element
- o Second largest element
- o Smallest element

```
➤ numbers = [12, 45, 7, 23, 56, 89, 34]
print("Largest element:",
max(numbers))numbers.sort(reverse=True)
print("Second largest element:", numbers[1])
print("Smallest element:", min(numbers))
```

- output :-

```
Largest element: 89
Second largest element: 56
Smallest element: 7
```

6. Create a list:

```
arr = [10, 20, 30, 40, 50, 60]
```

Write a function that takes the list as input and returns the list in reverse order without using the reverse() method.

```
➤ arr = [10, 20, 30, 40, 50, 60]
```

```
def reverse_list(lst):  
    return lst[::-1]
```

```
print("Original list:", arr)  
print("Reversed list:", reverse_list(arr))
```

- output :-

Original list: [10, 20, 30, 40, 50, 60]

Reversed list: [60, 50, 40, 30, 20, 10]

7. Create a tuple:

```
data = (5, 10, 15, 20, 25, 30, 35)
```

Write a program to:

- Count how many elements are divisible by 5
- Find the sum of all elements
- Find the average of the tuple

```
➤ data = (5, 10, 15, 20, 25, 30, 35)
```

```
count = 0  
for num in data:  
    if num % 5 == 0:  
        count += 1  
total = sum(data)  
average = total / len(data)
```

```
print("Count of elements divisible by 5:", count)  
print("Sum of all elements:", total)  
print("Average of the tuple:", average)
```

- output :-

Count of elements divisible by 5: 7

Sum of all elements: 140

Average of the tuple: 20.0

8. Create a dictionary:

```
students = { "Aman": 78, "Riya": 92, "Kriti": 88, "Rahul": 95 }
```

Write a program to:

- Find the student with the highest marks
- Find the student with the lowest marks
- Print only the students who scored more than 85 marks

```
➤ students = {  
    "Aman": 78,  
    "Riya": 92,  
    "Kriti": 88,  
    "Rahul": 95  
}
```

```
highest = max(students, key=students.get)
```

```
lowest = min(students, key=students.get)
```

```
print("Student with highest marks:", highest, "-",  
students[highest])
```

```
print("Student with lowest marks:", lowest, "-", students[lowest])
```

```
print("Students scoring more than 85 marks:")
```

```
for name, marks in students.items():
```

```
    if marks > 85:
```

```
        print(name, "-", marks)
```

- output :-

Student with highest marks: Rahul - 95

Student with lowest marks: Aman - 78

Students scoring more than 85 marks:

Riya - 92

Kriti - 88

Rahul – 95

9. Write a function:

`count_frequency(arr)`

that takes a list as input and prints the frequency of each element.

Example:

```
arr = [1, 2, 2, 3, 1, 4, 2]
```

Output:

1 -> 2 times

2 -> 3 times

3 -> 1 time

4 -> 1 time

```
➤ def count_frequency(arr):  
    freq = {}  
  
    for item in arr:  
        if item in freq:  
            freq[item] += 1  
        else:  
            freq[item] = 1  
  
    for key, value in freq.items():  
        if value == 1:  
            print(f"{key} -> {value} time")  
        else:  
            print(f"{key} -> {value} times")  
  
arr = [1, 2, 2, 3, 1, 4, 2]  
count_frequency(arr)
```

- output :-

1 -> 2 times

2 -> 3 times

3 -> 1 time

4 -> 1 time

10. Write a function:

`find_duplicates(arr)`

that takes a list as input and prints all duplicate elements present in the list.

Example:

`arr = [10, 20, 30, 20, 40, 10, 50, 30]`

Output:

Duplicate elements are:

10

20

30

```
➤ def find_duplicates(arr):  
    duplicates = set()  
    for item in arr:  
        if arr.count(item) > 1:  
            duplicates.add(item)  
  
    print("Duplicate elements are:")  
    for item in duplicates:  
        print(item)  
  
arr = [10, 20, 30, 20, 40, 10, 50, 30]  
find_duplicates(arr)
```

- output :-

Duplicate elements are:

10

20

30

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